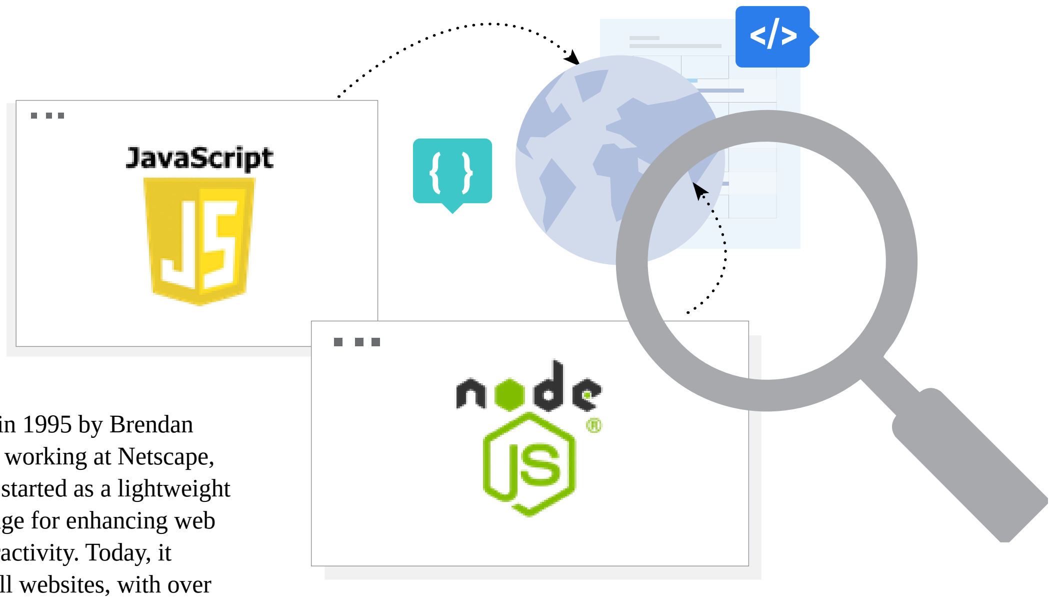
JavaScript and Node.js: How Open Source Fuelled the Web's Evolution

JavaScript is the lifeblood of the modern web. Node.js is a runtime environment that allows developers to run JavaScript on the server side. Both are not only relevant today but continue to lead the charge in modern web development. Let's find out how...



Introduced in 1995 by Brendan Eich while working at Netscape, JavaScript started as a lightweight scripting language for enhancing web pages with interactivity. Today, it powers nearly all websites, with over 97% of them using it for client-side functionality. Its impact stretches far beyond simple animations or form validations—it is used for full-scale application development, server-side scripting, real-time communication, game development, and even machine learning in the browser.

Its dynamic typing, prototype-based inheritance, and event-driven nature make it highly flexible, allowing developers to build rich user interfaces and interactive elements seamlessly. It integrates naturally with HTML and CSS, forming the foundation of front-end development. JavaScript engines like Google's V8 have significantly improved performance, turning JavaScript into a high-speed, modern programming language.

JavaScript's ability to run in any browser without needing installation, its constant evolution through ECMAScript standards, and the rise of powerful frameworks and tools have cemented its place as a cornerstone of web development. It has become so ubiquitous that any developer aiming to work on the web must know JavaScript. Its presence across devices and platforms—from desktops to mobiles, and even IoT—highlights its truly universal appeal.

Node.js: Revolutionising backend development with JavaScript

Before Node.js, JavaScript was confined to the browser. Back-end development requires other languages like PHP, Java, or Python. But with the introduction of Node.js in 2009 by Ryan Dahl, everything changed. Node.js is a runtime environment built on Chrome's V8 JavaScript engine that allows developers to run JavaScript on the server side.

What makes Node.js revolutionary is its non-blocking, event-driven architecture. Unlike traditional models where each client request blocks a thread, Node.js uses a single-threaded event loop that can handle thousands of concurrent connections efficiently. This makes it ideal for building real-time applications such as chat apps, live dashboards, and streaming platforms.

Node.js is highly scalable, performs exceptionally well under load, and is used by tech giants like Netflix, PayPal, Uber, and LinkedIn. Its npm (node